Research Article

Advik Jain (Student)
Pathways School Noida New Delhi India
Advik.jain1@pathways.in

Abstract
This paper describes a new way to sort emergency room patients using cameras and microphones. Standard emergency rooms face major delays because nurses have to physically check patient vitals and tie clinical data into computers by hand. To solve this bottleneck, this project introduces a tool that checks patient distress automatically within 60 seconds of their arrival into the hospital. The system uses pre-extracted text-based spreadsheets covering vital signals, breathing signals, and joint movement numbers. This setup avoids heavy audio or visual storage issues Still teaching the computer how to spot physical and vocal troubles faced by the patient. By merging these numbers the model learns to connect high heart rates and loud voices strain into a single score. This working prototype shows that machine learning can check multiple physical signs at once to deliver fast and reliable tracking. The system also eliminates manual typing delays, ensuring that high-risk patients are flagged for immediate care the moment they walk into a hospital waiting room without any traditional delays.

Keywords

Computer Science
Machine Learning
Software engineering
Emergency Department Triage

Introduction

Emergency room overcrowding is a serious problem because delays in treatment can put patients at risk. Normally, patients go through a triage process where nurses check their condition and assign an Emergency Severity Index (ESI) score from 1 (most urgent) to 5 (least urgent). However, this process takes time because nurses must manually record vital signs and type patient information, which can delay care and allow a patient's condition to worsen unnoticed. This research project aims to create and test an automatic emergency room triage system that can assess patients within sixty seconds using touchless methods. The system combines four types of information, including health records, voice patterns, breathing sounds, and body movement tracking, to quickly identify high risk patients and provide a faster and more complete assessment of their condition

Literature Review

This project is based on recent developments in computer vision, sound analysis, and medical datasets. Earlier studies tried to automate parts of emergency room triage, but many focused on only one symptom or required very powerful computers. Research by Shen and Guo showed that facial expressions can be used to measure pain levels accurately. Their system tracked small facial movements, such as narrowed eyes and lowered eyebrows, and matched doctor pain scores with 98.81% accuracy. However, the system needed a large and expensive computer, making it difficult to use in smaller devices such as ambulances.

Breathing sounds can also help doctors assess patients without physical contact. A recent study by a Research Collaboration Cohort developed a system that estimated breathing rates by listening to chest sounds and observing throat movement. The system produced highly accurate results, but it could not fully combine the audio and video information. Because of this limitation, loud voices or surrounding noise could interfere with the breathing analysis.

Medical datasets are also important for training artificial intelligence systems. The MIETIC dataset organized thousands of hospital records into emergency severity levels, while another study showed that machine learning models could predict which patients might require intensive care. However, these systems depended on nurses entering patient data manually. Research using body movement tracking also showed that cameras could identify walking problems and posture changes, but these systems only worked in controlled environments. This project addresses these limitations by combining information from health records, voice patterns, breathing sounds, and body movement into one fast and lightweight triage system.

Methodology

The framework of this prototype relies on a feature-level data fusion. It combines pre-extracted text based spreadsheets to simulate a real time camera and microphone setup. This strategy allows the model to learn the relationship between vital signs, breathing issues, and physical movements without requiring heavy video or audio storage.

Data Sourcing and Component Mapping

The prototype is trained by combining data from four public datasets into one single file:

Clinical Vitals and Text Layer: Patient information comes from the FedMML Emergency Department Triage Dataset. This dataset provides important values such as heart_rate, systolic_bp, diastolic_bp, and clinical_notes, which are linked to a target esi_level from 1 to 5.

Vocal Distress Layer: Voice related stress information comes from the Speech Emotion Detection Dataset. Instead of using raw audio recordings, this dataset provides already calculated values such as voice Pitch (Hz) and Speech_Rate (Words Per Minute).

Respiratory Distress Layer: Information about heavy breathing and lung sound patterns comes from the Coswara Heavy Cough Dataset. This dataset contains extracted audio features such as Mel frequency cepstral coefficients (MFCC_1 through MFCC_13) and zero_crossing_rates that represent breathing irregularities.

Biomechanical Layer: Body movement and posture changes are represented using a synthesized MediaPipe Pose coordinate dataset. This converts camera images into numerical values that track Gait Cadence, Stride Symmetry, and Trunk Lean Angle.

Preprocessing and Feature Fusion

Before training the classification model, a data cleaning process is carried out. First, slow laboratory measurements are removed from the hospital dataset because blood test results can take hours and are not useful for a quick sixty second patient assessment. Missing vital sign values are filled using median values based on patient age groups. For the speech dataset, emotional labels are converted into a simple binary safety value where "sad" and "angry" are marked as vocal distress (1), while "happy" and "neutral" are marked as normal speech (0).

For the camera tracking data, joint coordinates are adjusted based on the patient's shoulder width. This helps make sure that the patient's distance from the camera does not affect the movement measurements. Finally, standard scaling is applied to all audio features so that larger frequency values do not dominate smaller sound wave values. After the cleaning process is completed, all features are combined side by side using feature level fusion to create a final master training dataset.

Results and Discussions

(This area is yet to be completed as the model is still at the starting stages and is being coded)

Conclusion

This project creates a clear path towards a fast, touchless emergency room triage system by training a machine learning model on pre-extracted text features rather than raw video and audio streams. The prototype avoids heavy storage while keeping its diagnostic power at it’s peak. The final system will provide hospitals with an automated 60-second screening tool that identifies high-risk patients instantly upon their arrival into the hospital, removing human delays which may prevent the patients from receiving emergency care.

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During the preparation of this project Gemini was used to find a wide range of previous research articles and datasets which were then shortlisted by the author it was also used to polish certain sections of this article.After using this tool, the content was reviewed and edited as needed and the author takes full responsibility for the content of the publication.

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Author

Advik Jain Pathways School Noida New Delhi